Srivarshini Ksheerasagar

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SUMMARY

Graduate Researcher with 4+ years of experience building scalable **data systems** and **machine learning pipelines** using **Python**, **Apache Spark**, and **distributed computing frameworks**. Skilled in designing **ETL pipelines**, **data platforms**, and **analytics workflows** for large-scale **geospatial datasets**. Graduate Teaching Assistant for a senior-level **Software Testing** course, mentoring students in implementing **JUnit tests**, **debugging**, using **version control (GitHub)**, and applying **clean code** and **software engineering best practices**.

SKILLS

- **Programming Languages**: **Python**, **Java**, **C++**, **SQL**, **JSON**
- **Machine Learning & AI**: PyTorch (Advanced), CUDA, Transformers (HuggingFace), Scikit-learn
- **Data Engineering**: Apache Spark (Advanced), HDFS, MapReduce, DBT, ETL/ELT Pipelines
- **Cloud & Data Infrastructure**: AWS S3, Airflow, Snowflake, MongoDB Atlas, RBAC (Auth0)
- **Software Development**: Git/GitHub, Linux, JUnit Testing, Agile/Scrum, Jira
- **Tools & APIs**: MuleSoft, Anypoint Platform, REST APIs, Postman, Grafana, Power BI
- **Certifications**: **MuleSoft Certified Developer (MCD)**, **SnowPro Core (Snowflake)**

EXPERIENCE

Graduate Research Assistant | *Colorado State University* | Fort Collins, CO — Aug 2023 – Dec 2025

- Contributed to an **NSF-funded research project** by building **Python-based data pipelines** to collect, preprocess, and manage large-scale GOES-R satellite datasets, and designed a **Transformer-based model** in **PyTorch** to generate 24-hour satellite imagery predictions for downstream analytics.
- Engineered a **production-ready interactive UI platform** using **React**, **gRPC microservices**, and **Auth0 RBAC**, enabling **secure data access**, ingestion, and monitoring supporting ranching data across the U.S.
- Served as a **Graduate Teaching Assistant** for senior-level **Software Development** and **Big Data** courses, mentoring students on **MapReduce, Hadoop, HDFS, Spark, JUnit testing** and version control using **GitHub**.

Data Analyst | *Kipi.ai* | Houston, TX — Dec 2021 – Jul 2023

- Designed and maintained **end-to-end ELT pipelines** using **DBT** and **Snowflake**, and developed **optimized SQL models** and analytical datasets to support large-scale **data extraction**, **transformation**, and reporting workflows.
- Built **interactive dashboards** using **Tableau** and **Power BI**, collaborating with **Finance**, **Sales**, and **Product teams** to translate business metrics into **data-driven insights**, contributing to a **15% revenue increase**.

Software Intern | *Apisero(NTT Data)* | India — Aug 2021 – Dec 2021

- Designed and implemented **RESTful APIs** and **batch data pipelines** using **MuleSoft Anypoint Platform**, enabling scalable **data integration** and ingestion workflows.
- Implemented **watermark-based deduplication** for reliable processing and validated services through **REST endpoints** using **Postman**.

EDUCATION

Masters of Science in Computer Science, *Colorado State University* — Aug 2023 – Dec 2025

- **Coursework:** Machine Learning, Big Data, Distributed Systems, Software Testing

PROJECTS

Estimating Diurnal Patterns of Land Surface Temperature using Satellite Data (DayView)

- Designed **DayView**, a spatiotemporal **deep learning framework** based on a **Vision Transformer (ViT)** with a **Masked Autoencoder** strategy to reconstruct full 24-hour **Diurnal cycles** from sparse **GOES-R satellite** observations.
- Built scalable **ETL and preprocessing pipelines** for multi-terabyte satellite datasets, integrating **climatic zone and elevation features**, and conducted rigorous **model evaluation and empirical validation**.

Noble Research Institute – FFAR (Agricultural Data Analytics Platform)

- Led a team of six to design end-to-end **data pipelines** for multi-state agricultural datasets (OK, TX, CO, WY, MI).
- Developed a scalable **logical and physical data model** in MongoDB and implemented a secure real-time ingestion pipeline with Auth0 RBAC. Collaborated across multiple research institutions to deliver an extensible **data analytics platform**.

Land Use / Land Cover Classification using ResNet-50

- Developed a **deep learning based land cover classification model** using **ResNet-50** with **transfer learning** on **Sentinel-2 satellite imagery**, implementing extensive **data preprocessing** and **augmentation** to improve generalization and model robustness.
- Built an **end-to-end geospatial ML pipeline** for **data preprocessing, augmentation, model training, hyperparameter tuning, and evaluation** using **TorchVision**, **numPy** and **Google Earth Engine**, enabling classification and map generation for the state of California.

Building a Pastry Peer-to-Peer Network

- Implemented a **peer-to-peer distributed network** based on the **Pastry protocol** using **Java** and **Gradle**, building the system from scratch to enable decentralized node communication and efficient message routing.
- Designed core **network protocols** including node join, leave operations, routing tables, and message forwarding logic to support scalable and fault-tolerant **distributed system communication**.

Real-Time Sentiment Analysis Using Twitter Data

- Built a real-time **distributed data ingestion and analytics pipeline** using **Apache Spark Streaming** and **HDFS**, enabling live processing of Twitter data through a **TCP socket connection** between distributed nodes.
- Implemented sentiment classification and analytical modeling to identify **sentiment trends** and key metrics, delivering **scalable and reliable analytics** for data-driven insights.